Piyush Rajendra Nagpal

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Education

- **University of Colorado Boulder** — Boulder, CO
  Professional Master's in Embedded Systems Design; GPA: 3.77/4.00 — *Graduation: May 2, 2027*
- **International Institute of Information Technology (I2IT)** — Pune, India
  Bachelor of Technology – Electronics and Telecommunication; GPA: 8.7/10 — *2019 – 2023*

Technical Skills

- **Programming**: Python, C, C++, Bash, Dart, Java
- **Linux and Automation**: Linux, Python scripting, test automation, debugging, Git, Docker, Makefiles
- **Embedded Systems**: STM32, ESP32, Raspberry Pi, Arduino, embedded Linux, bare-metal firmware, RTOS concepts
- **Protocols**: UART, SPI, I2C, CAN, BLE, Wi-Fi, MQTT, gRPC, Protocol Buffers
- **Hardware and Tools**: PCB design, Altium Designer, oscilloscopes, logic analyzers, ADC/DAC, timers, DMA

Professional Experience

- **Qualcomm** — *Boulder, CO*
  Interim Engineering Intern – Infrastructure Test Team — *May 2026 – Aug 2026*
  - Developed Python-based test automation workflows to improve the efficiency and reliability of infrastructure testing.
  - Supported Linux-based testing environments through automation development, validation, and troubleshooting.
  - Collaborated with engineers to debug integration issues, streamline test execution, and improve visibility into test results.
- **Cognologix Technologies** — *Pune, India*
  Product Developer — *July 2023 – July 2025*
  - Developed embedded firmware and IoT product features for BLE-enabled systems, integrating microcontroller firmware, Flutter mobile applications, backend services, and hardware interfaces.
  - Implemented and debugged real-time communication over BLE, MQTT, and CAN, reducing field failures by 40% and improving device-to-cloud reliability.
  - Built and validated UART, SPI, I2C, GPIO, timer, and sensor-data interfaces across IoT deployments supporting more than 10,000 active devices.
- **University of Colorado Boulder** — *Boulder, CO*
  Graduate Learning Assistant – ECEN 5863 Programmable Logic Embedded System Design — *Aug 2026 – Present*
  - Support graduate-level digital logic coursework involving FPGA design, HDL, synthesis, timing analysis, and hardware validation.
  - Assist students with Quartus, ModelSim, Libero SoC, and FPGA development workflows through office hours and technical troubleshooting.
  - Evaluate digital design projects, review technical reports, and conduct project demonstrations and design reviews.

Research and Technical Teams

- **SWARM-EX CubeSat Project** — CU Boulder
  Volunteer – Command and Data Handling Team — *Oct 2026 – Present*
  - Contributing to Bluefin integration activities within the CubeSat Command and Data Handling subsystem, with emphasis on embedded systems and hardware-software integration.

Projects

- **AESD Rover Autonomy System – Yocto Linux, ESP32, CAN**: Built a distributed embedded rover using Raspberry Pi with Yocto Linux and ESP32 real-time motor/sensor control. Implemented SocketCAN communication, telemetry, heartbeat monitoring, watchdog-based fail-safe behavior, and autonomous control.
- **BLE-MQTT-CAN RV Awning Control System**: Developed an embedded Linux-based RV awning control prototype with Raspberry Pi as a BLE peripheral and MQTT gateway. Integrated mobile communication, CAN motor commands, and sensor telemetry.
- **Instrument Droid – 4-Layer Electronic Load System**: Designed a microcontroller-based electronic load featuring DAC-controlled MOSFET current sinking, ADC sampling, UART reporting, PCB layout, hardware bring-up, and calibration.

Honors

- Patron Award for Overall Academic Achievement; Copyright Registration ROC: L-123685/2023; Finalist – Elastic Run #Possibilities Hackathon; Finalist – Rakathon by Rakuten.